A12749W1

SECOND PUBLIC EXAMINATION

Honour School of Philosophy, Politics and Economics
Honour School of Economics and Management
Honour School of History and Economics

MICROECONOMICS

TRINITY TERM 2023

Wednesday 24 May, 14:30 – 17:30

There are TEN questions in this paper.

Answer ALL Part A questions and TWO Part B questions.

Part A attracts 40% of the total marks. Each question in Part A has the same weight. Each question in Part B attracts 30% of the total marks. For multipart questions, the weight assigned to each part is indicated in brackets.

Candidates may use a calculator as specified by the Department of Economics.

Do **not** turn over until told that you may do so.

Part A

1. Alesha likes music and goods. She has the utility function

$$u_A(x, y_A) = 8x - \frac{x^2}{2} + y_A$$

where x is the amount of music she plays and consumes and y_A is the quantity of goods she consumes. Alesha's income is 32. Bob dislikes hearing Alesha's music and has the utility function

$$u_B(x, y_B) = -\frac{x^2}{2} + y_B$$

where y_B is the quantity of goods he consumes. Bob's income is also 32. The price of goods is 1.

(a) Alesha controls x and bargaining between her and Bob is not feasible. What value of x does Alesha choose? What are the resulting utilities of Alesha and Bob? [20%]

(b) What is the Pareto efficient level of x? Explain how this is derived and why it is unique. [20%]

(c) What is the Pigouvian tax on Alesha's production of x that ensures that the Pareto efficient level is achieved? How much tax revenue is generated? [20%]

(d) The government wants to compensate Alesha with a monetary transfer financed by the tax revenue and Bob's income. How much compensation does Alesha need to be as well off with the tax as without it? Would compensating her with the tax revenue be enough? [40%]

2. There are two firms in an industry. Their goods are horizontally differentiated — they are imperfect substitutes. The demand function for firm a's good is

$$q_a(p_a, p_b) = 6 - p_a + 0.5p_b$$

and that for firm b's good is

$$q_b(p_a, p_b) = 6 - p_b + 0.5p_a$$

where p_a is the price of a's good and p_b is the price of b's good. The marginal cost of producing each good is 6. There is no fixed cost initially.

(a) Suppose first that firm b sets $p_b = 6$, so its price equals marginal cost. Explain why firm a will want to set its price above 6. [20%]

(b) Now suppose that each firm can choose any positive price, each seeks to maximize its profit, and prices are chosen simultaneously. Show that the Nash equilibrium in prices has each firm setting a price of 8. Calculate aggregate profits in the Nash equilibrium. [40%]

(c) The firms ask the competition authority for permission to merge. The merged firm will continue to produce both differentiated products and will set a single price, $p = p_a = p_b$, to maximize aggregate profit. The merger imposes a fixed cost of 20 on the merged firm but reduces the marginal cost of producing each good to 0. Would the competition authority allow the merger? Explain. [40%]

3. A risk neutral principal (P, she) is considering hiring an agent (A, he).

   P's revenue depends on A's effort, $e \in \{0, 1\}$. If A works hard ($e = 1$) then P's revenue is either 60 or 20 monetary units with a 50-50% chance. If A chooses $e = 0$ then P's revenue is 20 for sure, the same when A is not hired. P maximizes the expected value of her revenue minus the wage paid to A.

   A's cost of choosing $e = 1$ is 5 monetary units whereas $e = 0$ is costless for him. He maximizes the expected value of $u(x) = \sqrt{x}$ where x is his wage minus the cost of effort (if applicable). For instance, if A gets paid 30 monetary units and works hard then his utility is $u(30 - 5) = \sqrt{25} = 5$. If A faces risk then he takes the probability-weighted average of the utility of his net monetary payoff (wage minus effort cost).

   P offers a contract to A, which he either accepts or rejects. If A rejects then he gets slightly less than 9 monetary units from a third party with no effort. (In other words, A accepts any contract that yields an expected utility of 3 or more.)

   (a) Suppose that A's effort is contractible, that is, the wage offered by P to A can depend on the level of e chosen by A. What contract does P offer and what is her net profit? [20%]

   (b) From now on assume that A's choice of effort $e \in \{0, 1\}$ is not observable by P, but A's wage can be made contingent on P's revenue, which is called a revenue-sharing contract. Suppose P offers A a wage of 30 if her revenue is 60 and 6 if her revenue is 20. Does A accept or reject this offer, and in case he accepts, which effort level does he choose? What is P's net profit? [20%]

   (c) Find the revenue-sharing contract (two wage levels depending on P's revenue) maximizing P's expected profit under the assumption that e is not observable. Explain why A's participation and incentive constraints cannot hold with "slack" (i.e., as strict inequalities) in P's optimal contract. Compare the optimal contract with that seen in part (b). [40%]

   (d) Compute and interpret the agency cost in the context of this problem. [20%]

4. Cornelia's wealth consists of 2 million ducats at home and 8 million ducats abroad. She is an expected-utility maximizer with utility $u(w) = \ln(w)$ on final wealth w measured in millions of ducats. Cornelia knows that her wealth stored abroad will be confiscated with probability $p = \frac{1}{3}$ and wants to take out insurance against this adverse event.

Insurance companies are aware of Cornelia's initial wealth position and that her utility of final wealth is $\ln(w)$, but they are unsure about the probability that her foreign wealth will be confiscated. They think the probability of loss could be either $\frac{1}{3}$ or $\frac{1}{2}$. Currently they offer insurance against foreign wealth loss at premium $\pi = \frac{1}{2}$. That is, Cornelia can buy unit coverage (insurance that pays 1 ducat in case her wealth abroad is confiscated) for an up-front payment of $\pi = \frac{1}{2}$ ducats. She can buy any coverage at this unit price.

(a) Verify that Cornelia's final wealth when she buys x amount of coverage is $w_N = 10 - 0.5x$ in case her foreign wealth is not confiscated and it is $w_A = 2 + 0.5x$ in case her foreign wealth is confiscated. Write down her expected utility and find the level of coverage that she chooses. [30%]

(b) What would be the premium charged by competitive insurers if they knew that Cornelia's probability of loss was $\frac{1}{3}$ rather than $\frac{1}{2}$, and how much coverage would she buy in that case? [20%]

(c) The insurers, not knowing whether Cornelia's probability of loss is $\frac{1}{2}$ or $\frac{1}{3}$, offer her two options: either full coverage at premium $\pi = \frac{1}{2}$, or limited coverage $\bar{x}$ at premium $\bar{\pi} = \frac{1}{3}$. Under what condition on the value of $\bar{x}$ does this work as screening in insurance? Verify whether $\bar{x} = 2, 3$, or 4 satisfy the condition. [30%]

(d) Is Cornelia better off in any of the limited-coverage screening contracts with premium $\bar{\pi} = \frac{1}{3}$ found in (c), or by buying any coverage she likes at premium $\pi = \frac{1}{2}$? Which arrangement is more profitable for the insurers? Explain. [20%]

Part B

5. **EITHER**
   Do the fundamental theorems of welfare economics provide useful guides for public policy?

   OR
   A country that previously did not trade with the rest of the world opens itself to free trade. Outline the possible effects on production, consumption and the welfare of individuals in this country.

6. "The main problem with providing the efficient amount of a public good is that the government does not know how individuals value the good." Discuss.

7. How can the theory of repeated games be used to help to identify the factors that determine whether collusion in an industry is sustainable or not? Is it a problem that repeated games may have multiple equilibria?

8. How can expected utility theory be used to explain a risk-averse individual's decision to invest in a risky asset? Illustrate your answer using a model.

9. Since the 1990's Dutch traffic engineer Hans Monderman and his followers have been advocating for the removal of road markings and traffic control signs (e.g., centre lines, bicycle lanes, traffic lights) in order to make roads safer. Discuss the trade-offs from the perspective of moral hazard.

10. Consider a two-player simultaneous-move game with payoffs in expected utility units given in the table below. The payoff of player 1, choosing row, is written first and given by parameters t, x, y, and w. The values of t, x, y, and w are all different (distinct). Moreover, player 1's preferences satisfy the following two conditions:

(i) player 1 is indifferent between playing either T against L, or B against a 50-50% mixture of C and R;

(ii) player 1 is indifferent between playing either B against L, or T against a mixture of C and R where C is played with 80% and R with 20% chance.

	L	C	R
T	t, 5	x, 1	y, 2
B	w, 2	y, 6	x, 3

(a) Can some of the values of t, x, y, and w be set arbitrarily without imposing any additional conditions on player 1's preferences besides the expected-utility hypothesis (i.e., that she maximizes her expected payoff) and conditions (i) and (ii) above? For example, can we normalize $t = 5$ without affecting player 1's predicted behaviour? Can we normalize any other parameter values when $t = 5$? [25%]

(b) Assume $t = 5$. Find and eliminate all pure strategies that are strictly dominated for each player by any of their pure or mixed strategies. Repeat this procedure in as many rounds as possible. [10%]

(c) Assume $t = 5$, and in this part only, $x = 0$. Find all Nash equilibria of the game in pure and mixed strategies and compute the corresponding payoffs. Explain the method you use. [20%]

(d) Assume $t = 5$ and $x = 10$. Find all Nash equilibria of the game in pure and mixed strategies and compute the corresponding payoffs. Explain the method you use. [20%]

(e) Assume $t = 5$, $x = 10$, and that the game is repeated infinitely many times. Each player maximizes the discounted present value of his or her payoffs using discount factor $\delta \in (0, 1)$. At the beginning of each period the players publicly observe the outcome of a single coin flip, which is Heads or Tails with 50-50% chance (an independent draw every period). Is it possible to construct a Nash equilibrium in the infinitely-repeated game for δ sufficiently close to 1 in which the players play (T, L) when the coin flip comes up Heads and (B, R) when it comes up Tails? Explain. [25%]